FINAL YEAR PROJECT SPECIFICATION

K-Sync Malaysia

Database Design & Entity-Relationship Model

Chapter 3: System Architecture, Schema Definitions, and Relational Constraints

System Name	K-Sync Malaysia (Node.js & Express)
Scope Alignment	FYP Chapter 1: Introduction (10 Tables)
Production Database	Hostinger MySQL 8.0 / MariaDB
Development Database	SQLite 3 (better-sqlite3)
Flagship Model Entity	TXT World Tour in KL (Ref: KS-2026-TXT001)
Specification Version	Version 1.0 (Clean Scope)
Date Generated	October 2026

Prepared for Faculty Examination, Viva Defense, and Academic Archival

1. Database Architecture & Design Rationale

SECTION 1.0

The K-Sync Malaysia database is modeled specifically to solve the 3 core problems identified in **Chapter 1: Introduction**: (1) Fragmented event information, (2) Unverified submissions and scalper fraud, and (3) Disorganized community event coordination.

> **Design Optimization:** The schema was intentionally streamlined from 23 tables down to **10 purposeful tables**. Unnecessary overhead (such as separate venue lookup tables, GPS check-ins, and 1-on-1 private messaging) was eliminated, focusing 100% of system resources on centralized discovery, community discussion, and administrator verification.

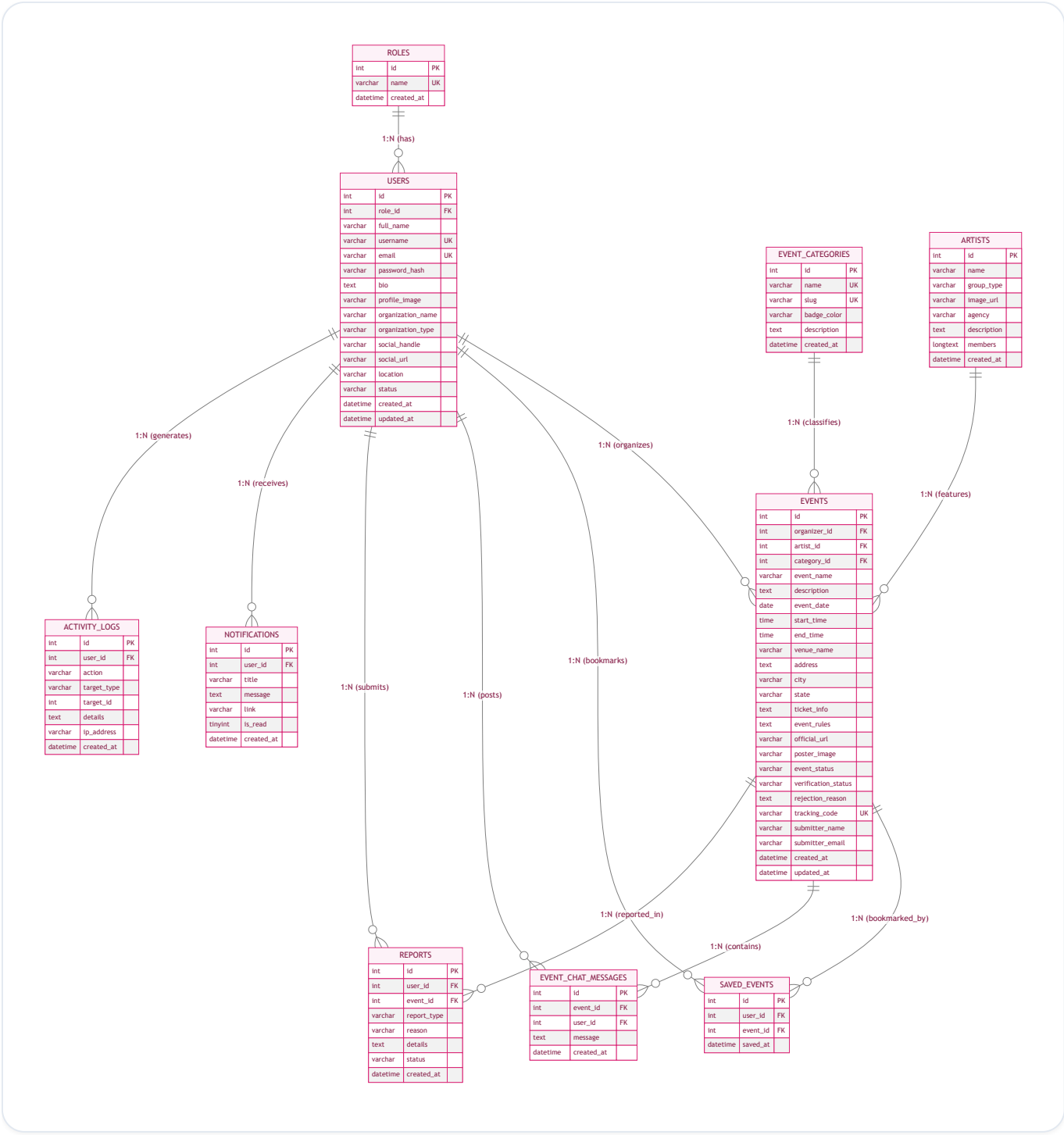
Key Structural Architectural Decisions:

- **Embedded Venue Attributes in Events:** Rather than maintaining a detached `venues` lookup table that requires complex relational joins, venue attributes (`venue_name`, `address`, `city`, `state`) are natively embedded into the `events` record. This enables seamless support for both mega arenas (Axiata Arena, Zepp KL) and custom fan cupsleeve cafes.
- **Single-Table Idol Member Biodata:** Group and soloist member biodata (stage name, birth name, positions, birthday, zodiac, height, blood type, and MBTI) is stored in the `artists.members` column formatted as structured JSON. This satisfies user requirements without introducing a redundant `artist_members` child table.
- **Dual Engine Compatibility:** The data dictionary supports both **Hostinger Cloud MySQL 8.0** in production (via `mysql2/promise`) and **SQLite 3** in local development, ensuring zero deployment friction.

2. Entity-Relationship Diagram (ERD)

SECTION 2.0

The diagram below illustrates all 10 entities, their primary/foreign key attributes, and relational multiplicities.


ROLES
int id PK
varchar name UK
datetime created_at
1:N (has)
USERS
int id PK
int role_id FK
varchar full_name
varchar username UK
varchar email UK
varchar password_hash
text bio
varchar profile_image
varchar organization_name
varchar organization_type
varchar social_handle
varchar social_url
varchar location
varchar status
datetime created_at
datetime updated_at
EVENT_CATEGORIES
int id PK
varchar name UK
varchar slug UK
varchar badge_color
text description
datetime created_at
ARTISTS
int id PK
varchar name
varchar group_type
varchar image_url
varchar agency
text description
longtext members
datetime created_at
1:N (generates)
1:N (receives)
1:N (organizes)
1:N (classifies)
1:N (features)
EVENTS
int id PK
int organizer_id FK
int artist_id FK
int category_id FK
varchar event_name
text description
date event_date
time start_time
time end_time
varchar venue_name
text address
varchar city
varchar state
text ticket_info
text event_rules
varchar official_url
varchar poster_image
varchar event_status
varchar verification_status
text rejection_reason
varchar tracking_code UK
varchar submitter_name
varchar submitter_email
datetime created_at
datetime updated_at
ACTIVITY_LOGS
int id PK
int user_id FK
varchar action
varchar target_type
int target_id
text details
varchar ip_address
datetime created_at
NOTIFICATIONS
int id PK
int user_id FK
varchar title
text message
varchar link
tinyint is_read
datetime created_at
1:N (submits)
1:N (posts)
1:N (bookmarks)
1:N (reported_in)
1:N (contains)
1:N (bookmarked_by)
REPORTS
int id PK
int user_id FK
int event_id FK
varchar report_type
varchar reason
text details
varchar status
datetime created_at
EVENT_CHAT_MESSAGES
int id PK
int event_id FK
int user_id FK
text message
datetime created_at
SAVED_EVENTS
int id PK
int user_id FK
int event_id FK
datetime saved_at

3. Cardinality & Multiplicity Relationship Matrix

SECTION 3.0

This table defines the participation constraints, multiplicities, foreign key mapping, and referential integrity delete actions across all relationships.

#	PARENT ENTITY	CHILD ENTITY	FOREIGN KEY COLUMN	CARDINALITY	BUSINESS RELATIONSHIP RULE	ON DELETE
1	**roles**	**users**	`users.role_id`	**1 : N**	One role is assigned to zero or many users. A user must have exactly 1 role.	`RESTRICT`
2	**event_categories**	**events**	`events.category_id`	**1 : N**	One category classifies zero or many events. An event belongs to exactly 1 category.	`RESTRICT`
3	**artists**	**events**	`events.artist_id`	**1 : N**	One artist performs in zero or many events. An event features zero or one artist.	`SET NULL`
4	**users**	**events**	`events.organizer_id`	**1 : N**	One organizer manages zero or many events. An event has 1 organizer account.	`RESTRICT`
5	**users & events**	**saved_events**	`user_id, event_id`	**M : N**	Many-to-Many resolved by associative table. Fans bookmark multiple events.	`CASCADE`
6	**events**	**event_chat_messages**	`event_id`	**1 : N**	One event hub contains zero or many community chat messages.	`CASCADE`
7	**users**	**event_chat_messages**	`user_id`	**1 : N**	One registered fan authors zero or many chat messages.	`CASCADE`
8	**events**	**reports**	`reports.event_id`	**1 : N**	One event listing can have zero or many content/scam reports.	`CASCADE`
9	**users**	**reports**	`reports.user_id`	**1 : N**	One user submits zero or many reports (optional if guest reports).	`SET NULL`
10	**users**	**activity_logs**	`activity_logs.user_id`	**1 : N**	One administrator/user triggers zero or many immutable audit records.	`SET NULL`

#	PARENT ENTITY	CHILD ENTITY	FOREIGN KEY COLUMN	CARDINALITY	BUSINESS RELATIONSHIP RULE	ON DELETE
11	**users**	**notifications**	`notifications.user_id`	**1 : N**	One user receives zero or many in-app alerts and soundcheck updates.	CASCADE

4. Complete Data Dictionary (10 Tables)

SECTION 4.0

4.1 Table: `roles`

RBAC Roles Table

Stores user role designations for role-based access control.

COLUMN NAME	DATA TYPE	NULL	KEY	DEFAULT	DESCRIPTION
id	`INT`	NO	PK	AUTO_INCREMENT	Surrogate primary key for role.
name	`VARCHAR(50)`	NO	UK	NULL	Unique role title: `Admin`, `Organizer`, `Fan`.
created_at	`DATETIME`	YES	-	CURRENT_TIMESTAMP	Timestamp when the role record was initialized.

4.2 Table: `users`

User Accounts & Profiles

Stores registered user credentials, profiles, and organizer verification metadata.

COLUMN NAME	DATA TYPE	NULL	KEY	DEFAULT	DESCRIPTION
id	`INT`	NO	PK	AUTO_INCREMENT	Unique user identifier.
role_id	`INT`	NO	FK	3	References `roles(id)`. Default 3 = Fan.
full_name	`VARCHAR(100)`	NO	-	NULL	User's full display or legal name.
username	`VARCHAR(50)`	NO	UK	NULL	Unique login username / handle.
email	`VARCHAR(100)`	NO	UK	NULL	Unique email address used for sign-in and notifications.
password_hash	`VARCHAR(255)`	NO	-	NULL	Bcrypt-hashed password string.
bio	`TEXT`	YES	-	NULL	Short user biography or fanclub description.
profile_image	`VARCHAR(255)`	YES	-	NULL	URL or uploaded path to avatar photo.
organization_name	`VARCHAR(150)`	YES	-	NULL	Official registered organizer or fanclub organization name.
organization_type	`VARCHAR(50)`	YES	-	'Fanclub'	Entity type: `Fanclub`, `Promoter`, `Agency`.
social_handle	`VARCHAR(50)`	YES	-	NULL	Official X / Instagram username (e.g. `@TXT_MY`).
social_url	`VARCHAR(255)`	YES	-	NULL	Direct URL to official organizer announcement channel.
location	`VARCHAR(100)`	YES	-	'Kuala Lumpur'	Malaysian city/state of the user.
status	`VARCHAR(20)`	YES	-	'ACTIVE'	Account status: `ACTIVE`, `SUSPENDED`.
created_at	`DATETIME`	YES	-	CURRENT_TIMESTAMP	User account registration timestamp.
updated_at	`DATETIME`	YES	-	CURRENT_TIMESTAMP	Timestamp of latest profile modification.

4.3 Table: `artists`

K-Pop Artists & Idol Biodata

Stores idol groups and soloists, embedding all member profiles directly inside JSON format.

COLUMN NAME	DATA TYPE	NULL	KEY	DEFAULT	DESCRIPTION
id	`INT`	NO	PK	AUTO_INCREMENT	Unique artist catalog identifier.
name	`VARCHAR(100)`	NO	-	NULL	Official artist/group name (e.g. `TOMORROW X TOGETHER`).
group_type	`VARCHAR(50)`	YES	-	'Boy Group'	Category: `Boy Group`, `Girl Group`, `Soloist`.
image_url	`VARCHAR(255)`	YES	-	NULL	Header / promotional group photo asset URL.
agency	`VARCHAR(100)`	YES	-	NULL	Management entertainment agency (e.g. `BIGHIT MUSIC`).
description	`TEXT`	YES	-	NULL	Career summary and background biography of the artist.
members	`LONGTEXT (JSON)`	YES	-	NULL	**JSON array holding individual idol biodatas** (Stage name, birth name, positions, DOB, zodiac, height, blood, MBTI).
created_at	`DATETIME`	YES	-	CURRENT_TIMESTAMP	Timestamp when artist was added to catalog.

4.4 Table: `event_categories`

Event Type Taxonomy

Categorization taxonomy for Malaysian K-Pop events.

COLUMN NAME	DATA TYPE	NULL	KEY	DEFAULT	DESCRIPTION
id	`INT`	NO	PK	AUTO_INCREMENT	Unique category identifier.
name	`VARCHAR(100)`	NO	UK	NULL	Category title (e.g. `Concert`, `Fan Meeting`, `Cupsleeve Event`).
slug	`VARCHAR(100)`	NO	UK	NULL	URL slug for filtering (e.g. `fan-meeting`).
badge_color	`VARCHAR(30)`	YES	-	'purple'	Tailwind color theme for UI badges (`purple`, `pink`, `rose`).
description	`TEXT`	YES	-	NULL	Scope and definition of the event category.
created_at	`DATETIME`	YES	-	CURRENT_TIMESTAMP	Timestamp when category was established.

4.5 Table: `events`

Centralized Event Information (Embedded Venue)

Central repository of verified K-Pop events, ticketing tiers, rules, embedded venue details, and public submission tracking codes.

COLUMN NAME	DATA TYPE	NULL	KEY	DEFAULT	DESCRIPTION
id	`INT`	NO	PK	AUTO_INCREMENT	Unique event identifier.
organizer_id	`INT`	NO	FK	2	References `users(id)` (Managing organizer/promoter).
artist_id	`INT`	YES	FK	NULL	References `artists(id)`. Set null if multiple artists.
category_id	`INT`	NO	FK	NULL	References `event_categories(id)`.
event_name	`VARCHAR(200)`	NO	-	NULL	Full official title of the Malaysian event.
description	`TEXT`	YES	-	NULL	Itinerary, setlist highlights, soundcheck perks, and overview.
event_date	`DATE`	NO	-	NULL	Calendar date of performance (`YYYY-MM-DD`).
start_time	`TIME`	NO	-	NULL	Doors open / concert start time (`HH:MM:SS`).
end_time	`TIME`	YES	-	NULL	Estimated conclusion time (used for 24h chat auto-lock).
venue_name	`VARCHAR(150)`	NO	-	NULL	**Embedded Venue Name** (e.g. `Zepp Kuala Lumpur`).
address	`TEXT`	YES	-	NULL	**Embedded Street Address** (e.g. `LaLaport BBCC, Jalan Hang Tuah`).
city	`VARCHAR(100)`	NO	-	NULL	**Embedded City** (e.g. `Kuala Lumpur`).
state	`VARCHAR(100)`	NO	-	NULL	**Embedded State** (e.g. `Wilayah Persekutuan Kuala Lumpur`).
ticket_info	`TEXT`	YES	-	NULL	Ticket tiers, seated categories, and pricing breakdown.
event_rules	`TEXT`	YES	-	NULL	Prohibited items (cameras, large bags), soundcheck queue rules.
official_url	`VARCHAR(255)`	YES	-	NULL	Official verified ticketing link (BookMyShow, GoLive, Ticket2U).
poster_image	`VARCHAR(255)`	YES	-	NULL	Promotional poster image URL or uploaded file location.
event_status	`VARCHAR(30)`	YES	-	'UPCOMING'	Life cycle: `UPCOMING`, `ONGOING`, `COMPLETED`.
verification_status	`VARCHAR(30)`	YES	-	'PENDING'	Review state: `PENDING`, `VERIFIED`, `REJECTED`.
rejection_reason	`TEXT`	YES	-	NULL	Administrator recorded reason explaining submission rejection.

COLUMN NAME	DATA TYPE	NULL	KEY	DEFAULT	DESCRIPTION
tracking_code	VARCHAR(50)	YES	UK	NULL	Unique tracking code (e.g. KS-2026-TXT001) for public status lookup.
submitter_name	VARCHAR(150)	YES	-	NULL	Name of the public submitter or fanclub organizer.
submitter_email	VARCHAR(150)	YES	-	NULL	Contact email where review notifications are delivered.
created_at	DATETIME	YES	-	CURRENT_TIMESTAMP	Submission timestamp.
updated_at	DATETIME	YES	-	CURRENT_TIMESTAMP	Last modification timestamp.

4.6 Table: saved_events

Fan Bookmarks Associative Table

Resolves the M:N relationship between users and bookmarked events with personal countdown timers.

COLUMN NAME	DATA TYPE	NULL	KEY	DEFAULT	DESCRIPTION
id	INT	NO	PK	AUTO_INCREMENT	Surrogate primary key.
user_id	INT	NO	FK	NULL	References users(id). Deleted on user cascade.
event_id	INT	NO	FK	NULL	References events(id). Deleted on event cascade.
saved_at	DATETIME	YES	-	CURRENT_TIMESTAMP	Timestamp when bookmark was saved.

4.7 Table: event_chat_messages

Live Event Discussion Hub

Stores per-event discussion messages exchanged between registered fans with automatic 24-hour post-event archival.

COLUMN NAME	DATA TYPE	NULL	KEY	DEFAULT	DESCRIPTION
id	INT	NO	PK	AUTO_INCREMENT	Unique chat message identifier.
event_id	INT	NO	FK	NULL	References events(id) to isolate discussions by show.
user_id	INT	NO	FK	NULL	References users(id) of message author.
message	TEXT	NO	-	NULL	Sanitized message body text.
created_at	DATETIME	YES	-	CURRENT_TIMESTAMP	Message creation timestamp.

4.8 Table: `reports`

Community Content Flagging

Records community reports flagging fraudulent listings, inaccurate event times, or inappropriate content.

COLUMN NAME	DATA TYPE	NULL	KEY	DEFAULT	DESCRIPTION
id	`INT`	NO	PK	AUTO_INCREMENT	Unique report tracking identifier.
user_id	`INT`	YES	FK	NULL	References `users(id)` (Reporter). Set NULL if anonymous guest.
event_id	`INT`	YES	FK	NULL	References `events(id)` being flagged.
report_type	`VARCHAR(50)`	YES	-	'EVENT'	Classification: `EVENT`, `CHAT`, `SCAM`.
reason	`VARCHAR(255)`	NO	-	NULL	Category: Inaccurate Date, Fake Event, Scalper Resale Link.
details	`TEXT`	YES	-	NULL	Supporting evidence or official link provided by reporter.
status	`VARCHAR(30)`	YES	-	'PENDING'	Resolution state: `PENDING`, `INVESTIGATING`, `RESOLVED`.
created_at	`DATETIME`	YES	-	CURRENT_TIMESTAMP	Filing timestamp.

4.9 Table: `activity_logs`

Administrator Audit Trail

Maintains an immutable historical record of all administrative verifications, approvals, and moderation actions.

COLUMN NAME	DATA TYPE	NULL	KEY	DEFAULT	DESCRIPTION
id	`INT`	NO	PK	AUTO_INCREMENT	Unique log identifier.
user_id	`INT`	YES	FK	NULL	References `users(id)` of the administrator who executed action.
action	`VARCHAR(50)`	NO	-	NULL	Action code: `APPROVE_EVENT`, `REJECT_EVENT`, `DELETE_USER`.
target_type	`VARCHAR(50)`	YES	-	NULL	Target entity: `events`, `users`, `categories`, `artists`.
target_id	`INT`	YES	-	NULL	Primary key of modified target record.
details	`TEXT`	YES	-	NULL	Human-readable summary of the administrative action.
ip_address	`VARCHAR(50)`	YES	-	NULL	Client IPv4/IPv6 address of executor for security auditing.
created_at	`DATETIME`	YES	-	CURRENT_TIMESTAMP	Action execution timestamp.

4.10 Table: `notifications`

User & Organizer Alerts

Stores user notifications for soundcheck queues, ticket sales, and event submission outcomes.

COLUMN NAME	DATA TYPE	NULL	KEY	DEFAULT	DESCRIPTION
id	`INT`	NO	PK	AUTO_INCREMENT	Unique notification identifier.
user_id	`INT`	NO	FK	NULL	References `users(id)` of recipient.
title	`VARCHAR(150)`	NO	-	NULL	Notification headline banner.
message	`TEXT`	NO	-	NULL	Detailed alert notification message.
link	`VARCHAR(255)`	YES	-	NULL	Target route URL (e.g. `/events/1`).
is_read	`TINYINT(1)`	YES	-	0	Read flag: `0 = Unread`, `1 = Read`.
created_at	`DATETIME`	YES	-	CURRENT_TIMESTAMP	Alert dispatch timestamp.

5. JSON Column Specification: `artists.members`

SECTION 5.0

To prevent schema bloat and preserve a lean 10-table architecture, idol group members are stored directly in the `artists` table inside the `members` column as a structured JSON array.

JSON Schema Definition:

```
{
  "$schema": "http://json-schema.org/draft-07/schema#",
  "title": "IdolMemberBiodataArray",
  "type": "array",
  "items": {
    "type": "object",
    "required": ["stage_name", "birth_name", "position", "dob"],
    "properties": {
      "stage_name":   { "type": "string", "description": "Public stage name (e.g. Yeonjun)" },
      "birth_name":   { "type": "string", "description": "Legal Korean/English birth name" },
      "position":     { "type": "string", "description": "Group role: Leader, Main Vocalist, Rapper, Dancer" },
      "dob":          { "type": "string", "description": "Date of Birth (e.g. September 13, 1999)" },
      "zodiac":       { "type": "string", "description": "Western astrological zodiac sign" },
      "height":       { "type": "string", "description": "Height in cm / feet-inches" },
      "blood_type":   { "type": "string", "description": "Blood type: A, B, AB, O" },
      "mbti":         { "type": "string", "description": "MBTI 4-letter personality type (e.g. ENFP)" },
      "nationality":  { "type": "string", "description": "Citizenship/country of origin" },
      "years_active": { "type": "string", "description": "Debut year to present" },
      "photo":        { "type": "string", "description": "Direct URL to official member profile portrait" }
    }
  }
}
```

Flagship Sample Payload: TOMORROW X TOGETHER (TXT)

```
[
  {
    "stage_name": "Yeonjun",
    "birth_name": "Choi Yeon-jun (최연준)",
    "position": "Main Rapper, Main Dancer, Vocalist",
    "dob": "September 13, 1999",
    "zodiac": "Virgo",
    "height": "181.5 cm / 5'11"",
    "blood_type": "A",
    "mbti": "ENFP",
    "nationality": "South Korean",
    "years_active": "2019–present",
    "photo": "https://images.unsplash.com/photo-1507003211169-0a1dd7228f2d?w=300"
  },
  {
    "stage_name": "Soobin",
    "birth_name": "Choi Soo-bin (최수빈)",
    "position": "Leader, Vocalist, Rapper",
    "dob": "December 5, 2000",
    "zodiac": "Sagittarius",
    "height": "185 cm / 6'1"",
    "blood_type": "A",
    "mbti": "ISFP",
    "nationality": "South Korean",
    "years_active": "2019–present",
    "photo": "https://images.unsplash.com/photo-1500648767791-00dcc994a43e?w=300"
  },
```

```
  {
    "stage_name": "Beomgyu",
    "birth_name": "Choi Beom-gyu (최범규)",
    "position": "Vocalist, Dancer, Rapper, Center, Visual",
    "dob": "March 13, 2001",
    "zodiac": "Pisces",
    "height": "180 cm / 5'11"",
    "blood_type": "AB",
    "mbti": "INFJ",
    "nationality": "South Korean",
    "years_active": "2019–present",
    "photo": "https://images.unsplash.com/photo-1519085360753-af0119f7cbe7?w=300"
  },
  {
    "stage_name": "Taehyun",
    "birth_name": "Kang Tae-hyun (강태현)",
    "position": "Main Vocalist, Dancer",
    "dob": "February 5, 2002",
    "zodiac": "Aquarius",
    "height": "177 cm / 5'10"",
    "blood_type": "A",
    "mbti": "ESTP",
    "nationality": "South Korean",
    "years_active": "2019–present",
    "photo": "https://images.unsplash.com/photo-1506794778202-cad84cf45f1d?w=300"
  },
  {
    "stage_name": "HueningKai",
    "birth_name": "Kai Kamal Huening (카이 카말 휴닝)",
    "position": "Main Vocalist, Dancer, Maknae",
    "dob": "August 14, 2002",
    "zodiac": "Leo",
    "height": "183 cm / 6'0"",
    "blood_type": "A",
    "mbti": "ISTP",
    "nationality": "Korean-American",
    "years_active": "2019–present",
    "photo": "https://images.unsplash.com/photo-1534528741775-53994a69daeb?w=300"
  }
]
```


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